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 Studierichting: Informatica
 Oefeningenreeks 5A nr. 11.5

$$\int_0^{\frac{\pi}{2}} e^{2x} \cos(3x) dx$$

$$\begin{cases} u = \cos(3x) \\ dv = e^{2x} dx \end{cases} \Rightarrow \begin{cases} du = -3 \sin(3x) dx \\ v = \frac{1}{2} e^{2x} \end{cases}$$

O.I.:

$$I = \frac{1}{2} e^{2x} \cos(3x) + \frac{3}{2} \int e^{2x} \sin(3x) dx$$

$$\begin{aligned} \begin{cases} u = \sin(3x) \\ dv = e^{2x} dx \end{cases} &\Rightarrow \begin{cases} du = 3 \cos(3x) dx \\ v = \frac{1}{2} e^{2x} \end{cases} \\ &= \frac{1}{2} e^{2x} \cos(3x) + \frac{3}{4} e^{2x} \sin(3x) - \frac{9}{4} \int e^{2x} \cos(3x) dx \end{aligned}$$

$$I = \frac{1}{2} e^{2x} \cos(3x) + \frac{3}{4} e^{2x} \sin(3x) - \frac{9}{4} I$$

$$I + \frac{9}{4} I = \frac{1}{2} e^{2x} \cos(3x) + \frac{3}{4} e^{2x} \sin(3x) + c$$

$$\frac{13}{4} I = e^{2x} \left(\frac{1}{2} \cos(3x) + \frac{3}{4} \sin(3x) \right) + c$$

$$I = \frac{4}{13} e^{2x} \left(\frac{1}{2} \cos(3x) + \frac{3}{4} \sin(3x) \right) + c$$

B.I.:

$$\begin{aligned} &\frac{4}{13} e^{\pi} \left(\frac{1}{2} \cdot 0 + \frac{3}{4} \cdot (-1) \right) - \frac{4}{13} \cdot 1 \left(\frac{1}{2} \cdot 1 + 0 \right) \\ &= \frac{-3e^{\pi}}{13} - \frac{2}{13} \\ &= \frac{-3e^{\pi} - 2}{13} \end{aligned}$$

References